

$$\text{KED}_{N_\alpha}(=\text{PED}_{N_\alpha}) = \frac{K_{N_\alpha}}{\sum K_{N_\alpha}} = \frac{m_N \boldsymbol{r}_{N_\alpha}^2}{\sum m_j \boldsymbol{r}_{j\alpha}^2} \quad (1)$$